INDIAN INSTITUTE OF TECHNOLOGY KANPUR

ME630 :: COMPUTATIONAL FLUID DYNAMICS

ASSIGNMENT 3

Unsteady SIMPLE Solver

Verification, SIMPLEC/SIMPLER and Central Differencing

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NOTE: Due to max page limitations, not all plots are added to this report. But all solutions and simulation runs are added as zip files. Also, For Case D, Re = 400 and 800 were not resolved for redundancy and the data was collected from previous questions.

Part A1 :: Developing channel flow (Re = 100)

The channel case was solved using a staggered-grid FVM, implicit backward-Euler stepping, UDS convection and SIMPLE using first order upwinding scheme.

A1.1 :: Outlet profiles and centreline

t (s)	1	2	5	10	19.25 (steady)
max $\|u - u_e\|/U_0$	0.240	0.124	0.0151	0.00156	0.00138

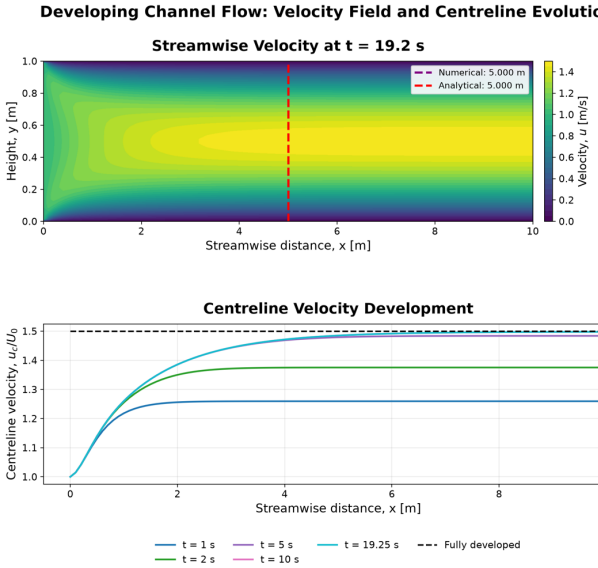


Fig. A1.1. Developing streamwise velocity and centreline evolution; archived Nx = 100, Ny = 40 run.

A1.2–A1.3 :: Entrance length and grid refinement

The entrance length is taken as the first x-location where u_c ≥ 0.99(1.5U_0) = 1.485 m/s. The empirical estimate is $L_e \approx 0.05\,Re\,H = 5.0$ m.

Grid	Δx	Δy	L_e (m)	Outlet error
100 × 20	0.100	0.050	5.90	0.00423
200 × 20	0.050	0.050	5.80	0.00417
100 × 40	0.100	0.025	5.00	0.00138

Halving Δx reduces L_e by 0.10 m, whereas halving Δy reduces it by 0.90 m.

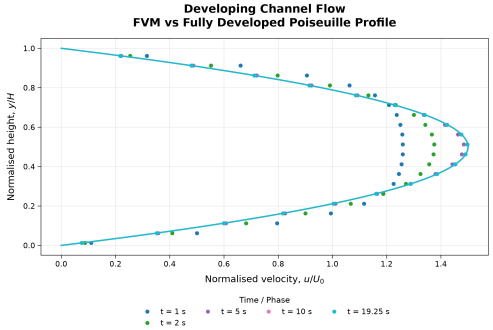
Wall-normal refinement has a larger effect on the outlet error because it resolves the near wall velocity gradients better.

The streamwise cell Péclet number $Pe_x = U_0\Delta x/\nu$ is 10 and 5 for Δx = 0.1 and 0.05 m. Estimated upwind false diffusion, $U_0\Delta x/2$, gives $\nu_{num}/\nu \approx Pe_x/2 = 5$ and 2.5, respectively.

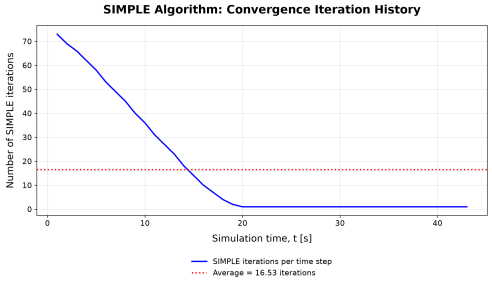
A1.4 :: Time-step independence of the steady state

Δt (s)	Steady t (s)	Outlet error	CPU (s)	Mean SIMPLE/step
0.10	14.50	0.00138	18.194	9.12
0.25	19.25	0.00138	13.381	12.52
1.00	43.00	0.00138	9.762	16.53

All time steps give the same final outlet error. However, the reported time to reach steady flow changes with reduction in Δt because the stopping condition compares preceeding time levels. Larger time steps require more iterations per time step but fewer time steps overall.



A1(a) Velocity profiles, 100×40, Δt = 0.25 s



A1(b) SIMPLE iterations, 100×40, Δt = 1.00 s

Part A2 :: Stokes first problem

The Stokes first problem uses a lower wall moving at U_0 = 1 m/s, with ν = 0.01 m²/s, H = 1 m and Ny = 200. The exact Similarity solution is u/U_0 = erfc(η), where η = y/(2√νt). The computed profiles do collapse onto a single curve in similarity coordinates as shown in the following plots.

A2.1–A2.3 :: Analytical comparison, similarity and wall shear

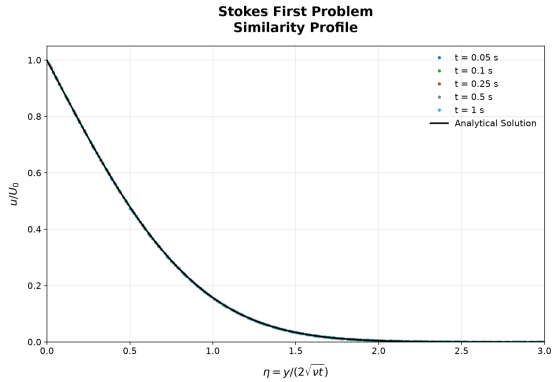


Fig. A2.1. Numerical profiles plotted against η; comparison with erfc(η).

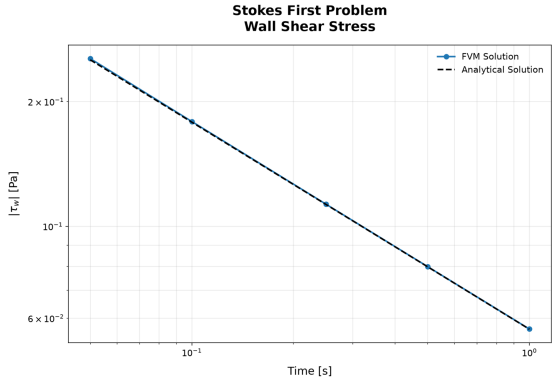


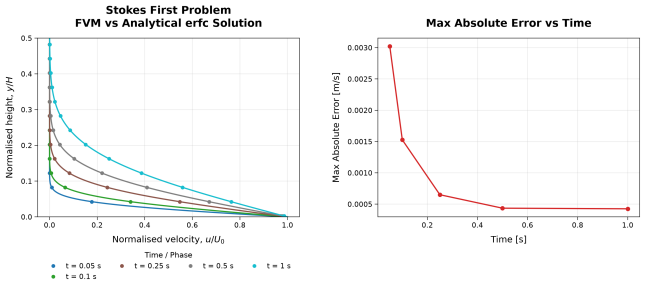
Fig. A2.2. Wall-shear magnitude from the solution and exact $t^{-1/2}$ variation.

Wall shear is estimated as $\tau_{w,num} \approx \mu[u_1 - U_0]/(\Delta y/2)$ and compared with $\tau_{w,exact} = -\mu U_0/\sqrt{(\pi\nu t)}$. The expected slope on log–log axes is −1/2. For Δt = 0.0005 s, the archived profile error decreases from 3.02×10^{-3} at t = 0.05 s to 4.26×10^{-4} at t = 1 s.

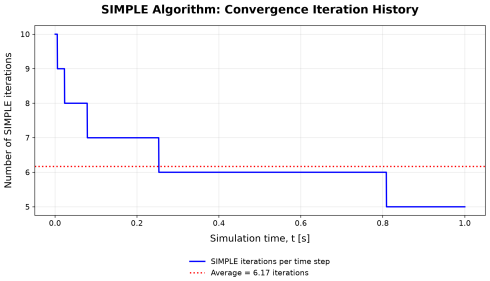
A2.4–A2.5 :: Time-step error and far-field boundary

Δt (s)	0.004	0.002	0.001	0.0005
t sampled (s)	0.248	0.250	0.250	0.250
max error/U_0	0.00260	0.00148	0.000924	0.000651

The measured log–log error slope is about 0.67, below the formal first-order time accuracy of backward Euler. The fixed spatial error and the sample at 0.248 s may have affected this fit. Grid-refinement and reduced-H results are not available in the supplied logs. At t = 1 s, $\delta_{99} = 3.64\sqrt{(\nu t)} \approx 0.364$ m, whereas H = 1 m. The domain-height limit at which the far boundary starts affecting the profile has not been tested numerically.



A2(a) Velocity profiles, 4×200, Δt = 0.0005 s



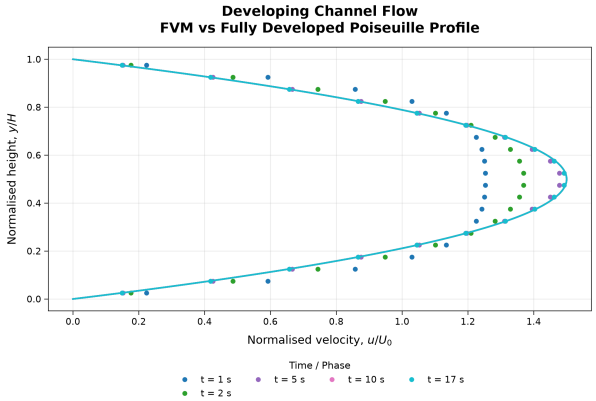
A2(b) SIMPLE iterations, 4×200, Δt = 0.0005 s

NOTE: Answers common to Part A and Part A3.5 are together noted down after Part A3.5

Part A :: Additional verification plots

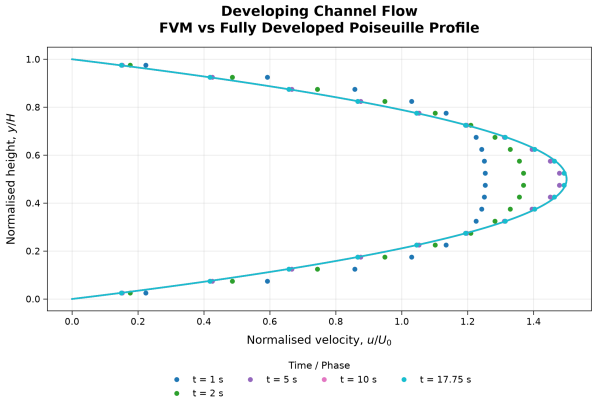
Grid refinement and time-step comparison

A-V1 | Re = 100, 100×20, Δt = 0.25 s



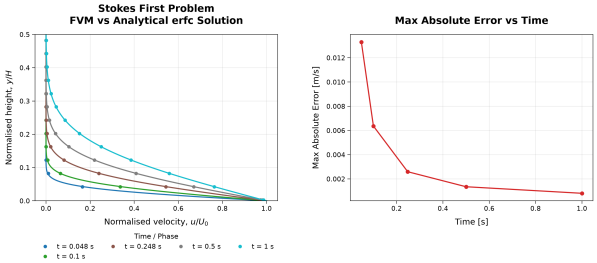
Coarse channel profile: effect of wall-normal resolution.

A-V2 | Re = 100, 200×20, Δt = 0.25 s



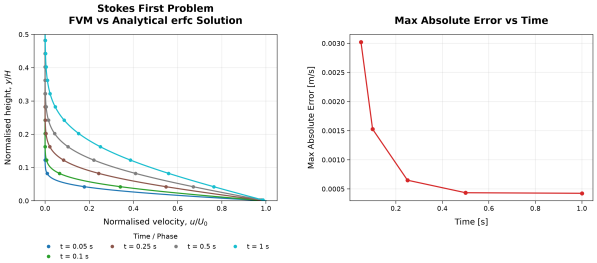
Same wall-normal grid with halved streamwise spacing.

A-V3 | Stokes I, Δt = 0.004 s



Coarse-step transient solution versus analytical profile.

A-V4 | Stokes I, Δt = 0.0005 s



Refined time-step comparison at fixed 4×200 grid.

Part A3 :: Stokes second problem

For Stokes second problem, the plate oscillates with velocity $U_0 \cos \omega t$, where $\omega = 2\pi\ s^{-1}$ and $\nu = 0.01\ m^2/s$. The exact periodic solution is $u/U_0 = e^{-ky} \cos(\omega t - ky)$, with $k = \sqrt{[\omega/(2\nu)]} = 17.7245\ m^{-1}$ and $\delta = 1/k = 0.056419$ m. The analytical envelopes are $\pm e^{-ky}$.

A3.1–A3.3 :: Eight phases and wall shear

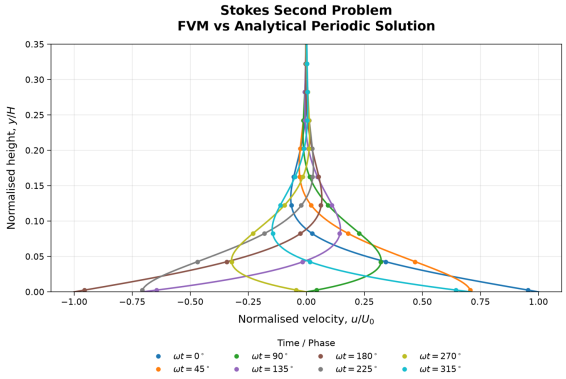


Fig. A3.1. Eight-phase velocity profiles at 400 steps/period with analytical profiles.

For the wall-fluid sign convention, the shear is $\tau_fluid = -\sqrt{2}\ \mu k U_0 \cos(\omega t + \pi/4)$, while the opposite plate-on-fluid traction is $+\sqrt{2}\ \mu k U_0 \cos(\omega t + \pi/4)$. Here, the traction exerted by the plate on the fluid leads the plate velocity by 45°, whereas the fluid-on-wall shear has the opposite sign. The plotted analytical shear curve is available in the diagnostics; a numerical phase-fit of the shear was not logged.

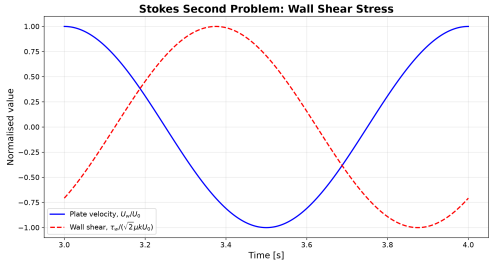
A3.4 :: Temporal accuracy of amplitude and phase at y = δ

Steps/T	Δt (s)	Amplitude	Error (%)	Lag (rad)	|Δlag| (rad)
50	0.0200	0.356447	3.1075	0.967626	0.032374
100	0.0100	0.361879	1.6311	0.983992	0.016008
200	0.0050	0.364684	0.8687	0.992021	0.007979
400	0.0025	0.366111	0.4808	0.995982	0.004018

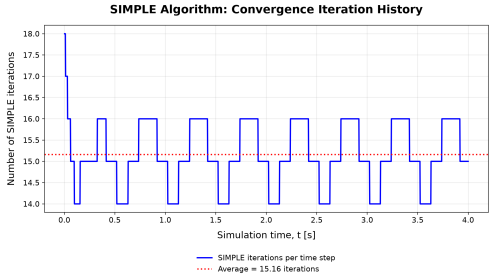
The fitted amplitude and phase are compared with $A = e^{-1} = 0.367879$ and phase lag = 1 rad. For backward Euler applied to $\dot{q} = i\omega q$, the amplification factor is $G = (1 - i\omega\Delta t)^{-1}$. Since $|G| < 1$, the method introduces numerical damping and phase error. Both errors decrease approximately by half when Δt is halved.

A3.5 and common Part A questions

For Ny = 200, δ/Δy = 11.28 and the amplitude error is below 1% for at least 200 steps per period. This establishes that the resolution is sufficient, but not the minimum required. In streamwise-uniform periodic Stokes flow, continuity gives vN = vS; impermeable boundaries imply v = 0, and transverse momentum gives uniform pressure. These tests check the transient term, viscous diffusion and wall boundary conditions, but not general 2D convection or pressure–velocity coupling. In Stokes I, the velocity change decreases with time, providing a better initial guess for each SIMPLE step. In Stokes II, the iteration count is mainly affected by αu = 0.8 and the 10^{-6} tolerance.



A3(a) Plate velocity & wall shear, 4×200, Δt = 0.0025 s

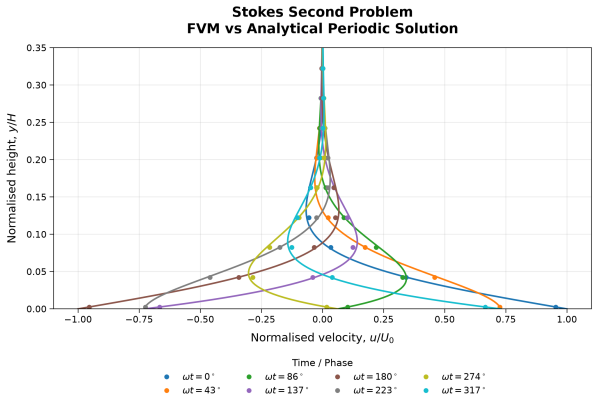


A3(b) SIMPLE iterations, 4×200, Δt = 0.0025 s

Part A3 :: Oscillatory-flow resolution study

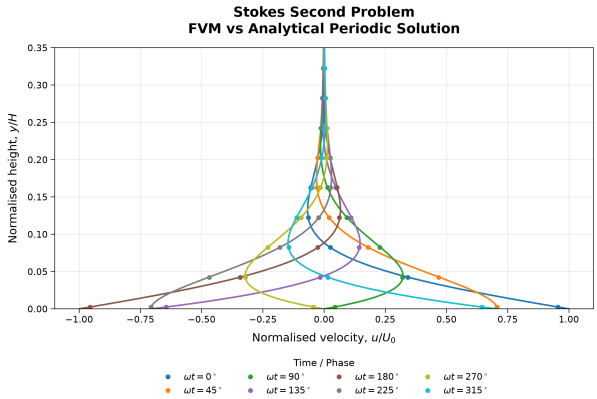
Stokes second problem: dependence on time-step resolution

A3-V1 | Stokes II, 50 steps/period



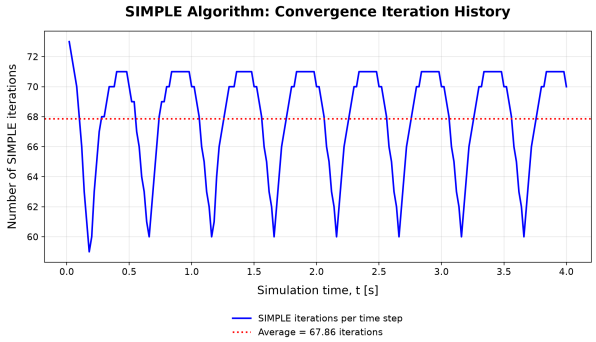
Coarse temporal resolution, Δt = 0.02 s.

A3-V2 | Stokes II, 400 steps/period



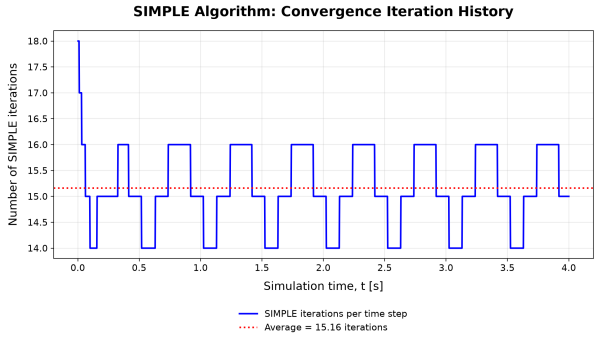
Refined temporal solution, Δt = 0.0025 s.

A3-V3 | Stokes II, 50 steps/period



Inner-iteration history at coarse time resolution.

A3-V4 | Stokes II, 400 steps/period



Inner-iteration history at refined time resolution.

Part B :: Backward-facing step

The BFS case uses S = 1, h = 17/16, H = S + h and L = 40S, with Re = $\rho U_0(2h)/\mu$. A parabolic inlet profile is prescribed above the step lip and zero velocity below it, with no-slip walls and a zero-gradient outlet. The reported Part B runs use SIMPLE with UDS on a 320 × 33 grid.

B1–B2 :: Recirculation and location extraction

Wall separation and reattachment are identified from changes in sign of near-wall velocity. Linear interpolation gives $x_{zero} = x_i - u_i(x_{i+1} - x_i)/(u_{i+1} - u_i)$. At Re = 100, a primary recirculation region forms behind the step. No upper-wall bubble is detected.

Re	x_1/S	x_4/S	x_5/S
100	2.9168	—	—
400	7.3811	—	—
800	8.2957	6.4541	13.8641

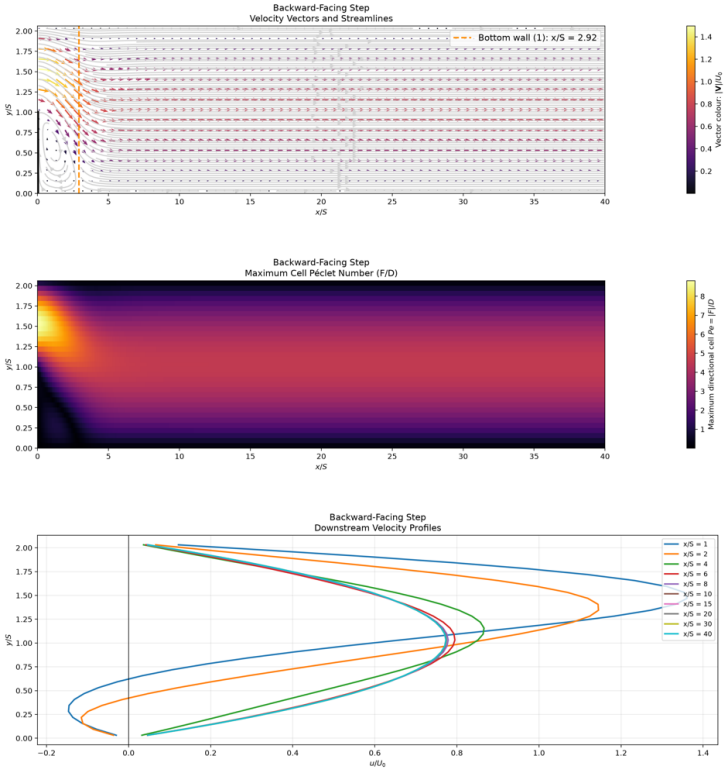


Fig. B1. Archived Re = 100 BFS flow field and near-wall recirculation diagnostics.

B3 :: Time-step sensitivity and failures

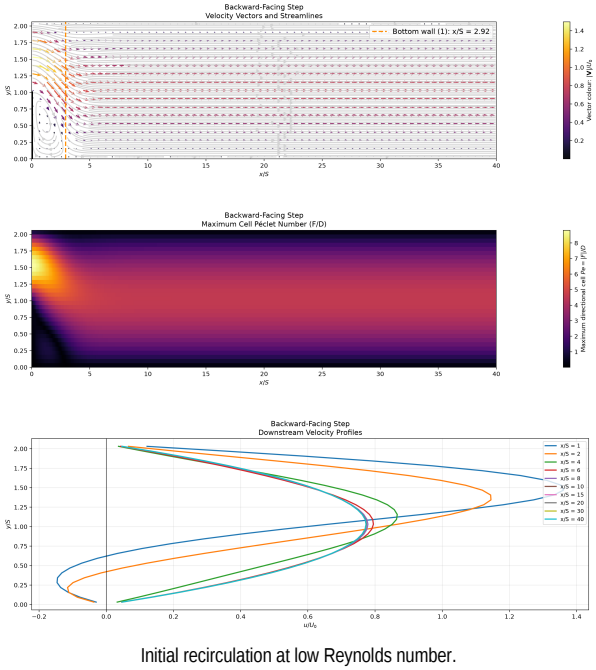
Re	Δt (s)	Mean iterations/step	x_1/S	Status
400	0.05	6.46	7.3811	Completed
400	0.50	17.66	7.3804	Completed
400	1.00	27.16	7.3789	Completed
800	0.05	7.60	8.2957	Completed
800	0.10	10.36	8.2973	Completed
800	0.30	15.96	8.3011	Completed
800	0.50 / 0.90 / 1.0	—	—	Inner nonconvergence

For Re = 800, runs with large time steps repeatedly reach the maximum SIMPLE iteration count, even when the continuity residual is very small. The velocity-change criterion remains large, sometimes $O(10^2–10^3)$, showing poor convergence of the inner iterations. Re = 400 completed for tested steps up to Δt = 1 s, while Re = 800 completed up to Δt = 0.3 s in this set. These are the largest successful tested steps, not proven stability limits. Strict steady-state convergence was not reached by t = 100 s for Re = 400 or 800.

Part B–C :: Additional flow and convergence plots

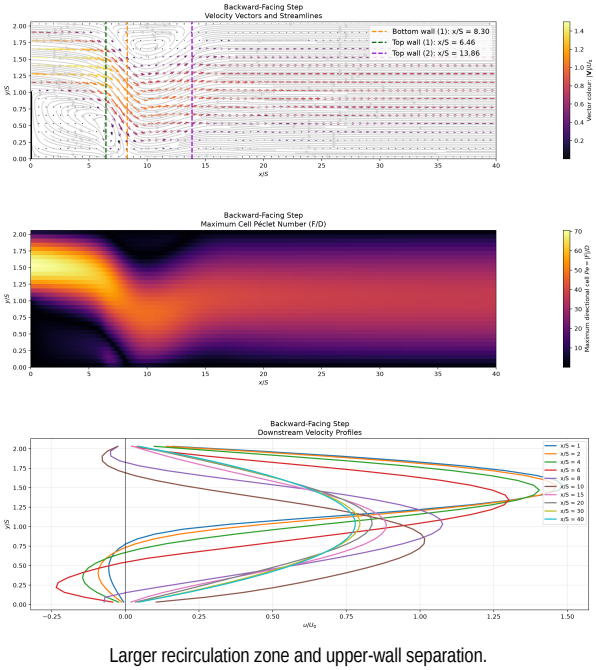
Flow structure and pressure–velocity coupling

BC-V1 | BFS Re = 100, SIMPLE–UDS



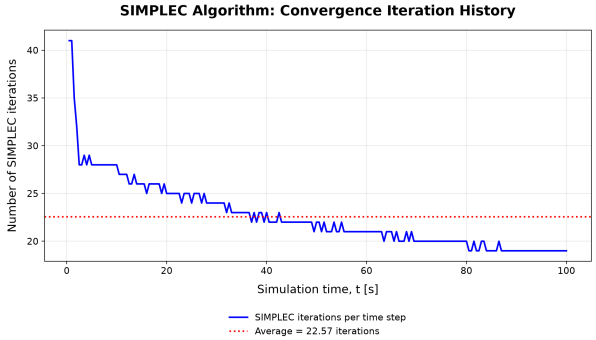
Initial recirculation at low Reynolds number.

BC-V2 | BFS Re = 800, SIMPLE–UDS



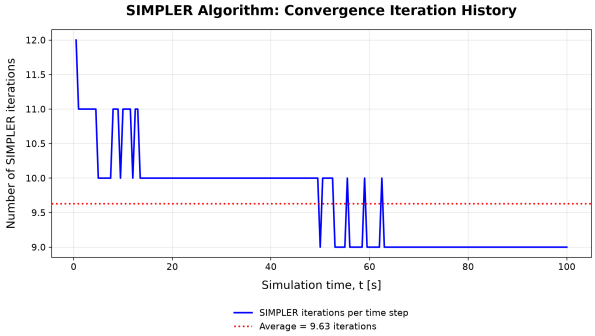
Larger recirculation zone and upper-wall separation.

BC-V3 | Re = 800, SIMPLEC, Δt = 0.5 s



Inner-iteration history using SIMPLEC.

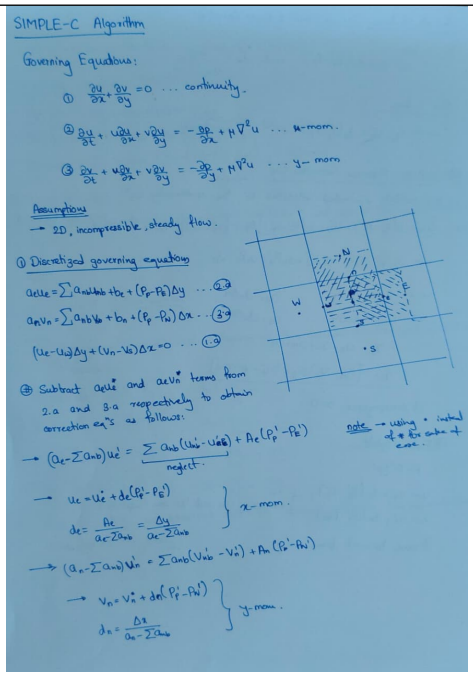
BC-V4 | Re = 800, SIMPLER, Δt = 0.5 s

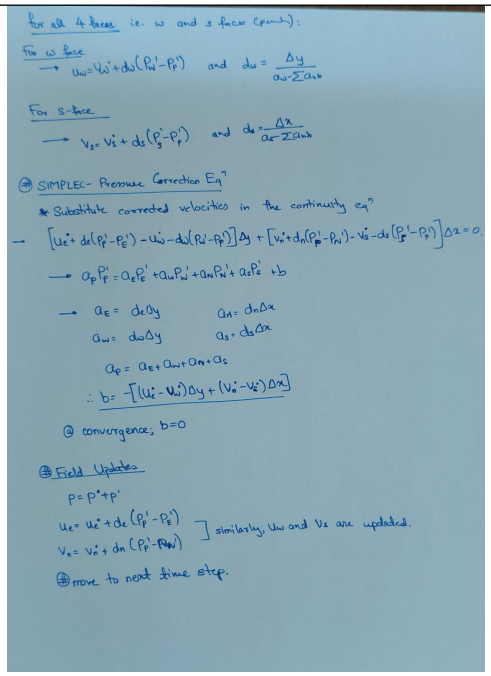


Iteration history for the same BFS case with SIMPLER.

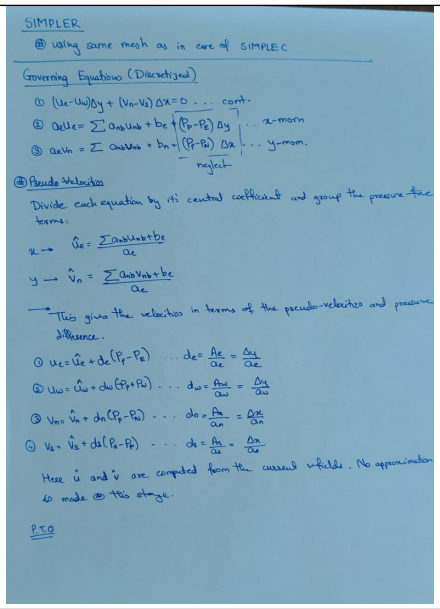
Part C1–C2 :: SIMPLEC and SIMPLER: derivation

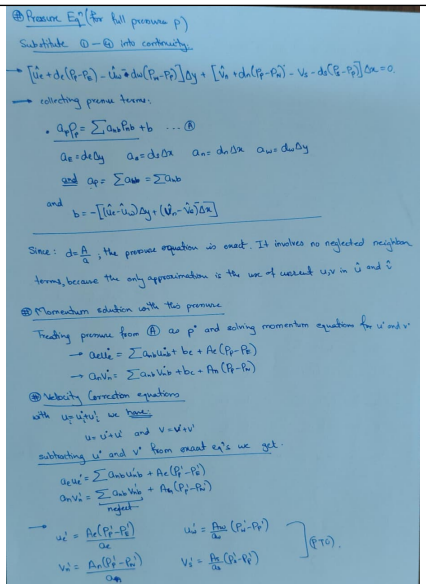
C1 :: SIMPLEC

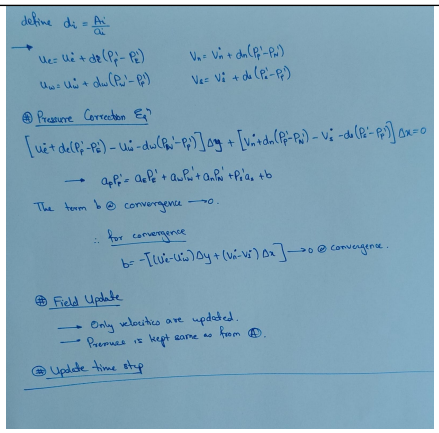




C2 :: SIMPLER







Advantages with increasing dt

The transient diagonal coefficient is $aP^0 = \rho\Delta V/\Delta t$. At small Δt, aP^0 dominates and neighbour correction coupling is weak, so all three schemes behave similarly. At large Δt, aP^0 shrinks; neglecting neighbour corrections in SIMPLE becomes a larger approximation, while SIMPLEC/SIMPLER better handle momentum–pressure coupling.

Distinguishing test case:

For periodic Stokes flow, $v \simeq 0$ and pressure is uniform, so pressure correction has little role. The backward-facing step has reverse flow, two-dimensional pressure gradients and recirculation. It is therefore a more useful test of pressure–velocity coupling algorithms.

Part C3 :: Algorithm performance and solver consistency

C3.1 :: Stokes II (200 steps per period; Ny = 200)

αu	Algorithm	Mean iter./step	CPU (s)	Amplitude at δ
0.5	SIMPLE	67.34	81.90	0.364571
0.5	SIMPLEC	67.34	82.50	0.364571
0.5	SIMPLER	67.34	153.40	0.364571
0.8	SIMPLE	23.36	29.01	0.364684
0.8	SIMPLEC	23.36	31.67	0.364684
0.8	SIMPLER	23.36	57.79	0.364684
1	SIMPLE	Failed	—	—
1	SIMPLEC	2.00	2.71	0.364716
1	SIMPLER	2.00	4.69	0.364716

For αu = 0.5 and 0.8, the three algorithms show the same iteration counts and nearly identical amplitudes at each fixed αu. SIMPLER takes more CPU time because it involves an additional pressure solve. At αu = 1, SIMPLE fails due to non-finite values in the five-point system, whereas SIMPLEC and SIMPLER converge in 2 iterations per step. The SIMPLE run at α = 0.3 requested in the assignment is not available in the archives.

C3.2 :: BFS tests (330 × 33, Δt = 0.5 s, UDS)

Re	Algorithm	Mean iter.	CPU (s)	x_1/S	x_4/S	x_5/S
400	SIMPLE	47.88	542.2	7.277	6.506	8.559
400	SIMPLEC	19.74	252.7	7.278	6.506	8.559
400	SIMPLER	7.11	214.2	7.278	6.506	8.559
800	SIMPLE	52.41	652.8	7.980	6.162	13.508
800	SIMPLEC	22.57	322.6	7.980	6.162	13.508
800	SIMPLER	9.63	265.0	7.980	6.162	13.508

At Re = 800, SIMPLER reduces the mean iteration count from 52.41 to 9.63 and the CPU time from 652.8 to 265.0 s, a reduction of about 59.4%. These timings are for simulations up to 100 s, not verified times to reach steady state. All three methods produce almost identical reattachment and separation locations. The successful Δt = 0.5 s run does not establish the maximum stable time step.

C3.3 :: Iteration histories (Re = 800)

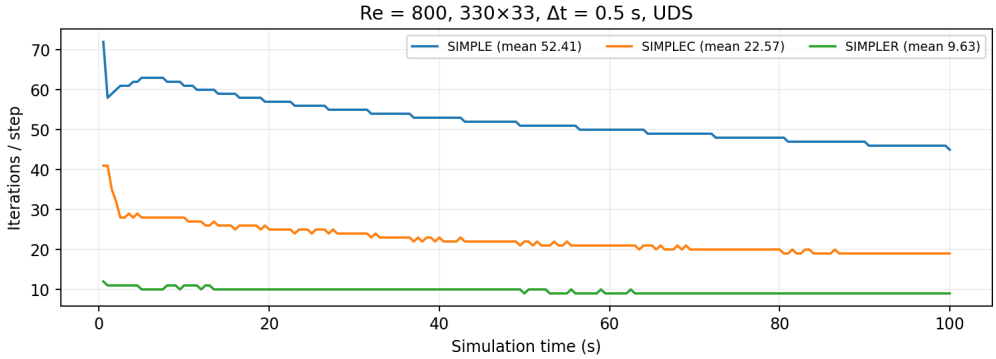


Fig. C3. Convergence iterations per time step from the three archived CSV histories.

Part D1–D2 :: Central differencing and deferred correction

The Part D BFS results are reported for Re = 400, a 340 × 33 grid and Δt = 0.05 s using SIMPLE. Although the assignment specifies SIMPLEC for Part D, the supplied Part D diagnostics identify SIMPLE. The comparison therefore tests the convection schemes under SIMPLE, not SIMPLEC.

D1 :: Péclet criterion and boundedness

For the transported velocity component the direct-CDS neighbour coefficient at the east face is aE = De − Fe/2 (similarly aW = Dw + Fw/2, etc.). When |Pe| = |F|/D > 2 in the outgoing-flow direction, a neighbour coefficient can become negative. This breaks the positive-neighbour-coefficient condition for monotonicity and can violate the Scarborough sufficient convergence/boundedness criterion. Oscillations are possible, although not compulsory in every run.

Scheme	Max cell Pe	Cells with Pe > 2	Fraction	x_1/S	Mean it./step
UDS	37.354	10,213 / 11,220	91.0%	7.3076	12.09
CDS	37.354	10,046 / 11,220	89.5%	7.9651	15.10
CDS-DC	37.354	10,046 / 11,220	89.5%	7.9651	12.91

The CDS and CDS-DC runs both complete and produce almost the same $x_1/S = 7.9651$. However, the logs do not contain a numerical measure of oscillations or the number of negative matrix coefficients. Hence the presence or magnitude of wiggles cannot be confirmed from these logs alone.

Interpretation of UDS and CDS-DC results: First-order upwinding introduces artificial diffusion, approximately |U|Δx/2 in a simple one-dimensional transport estimate. At large cell Péclet numbers, this numerical diffusivity may be much greater than the physical viscosity. It smooths local maxima and minima, weakens reverse-flow peaks and reduces the sharpness of the velocity field. A converged UDS solution may therefore have a noticeable discretisation error. UDS and converged CDS-DC do not solve exactly the same convection discretisation, so their reattachment lengths need not agree even when both runs converge.

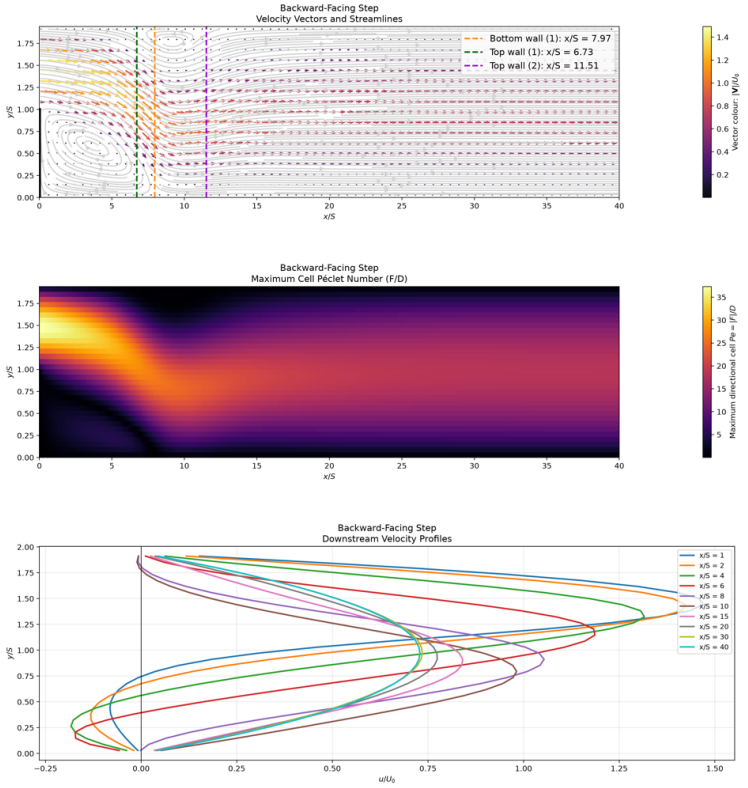


Fig. D1. Representative CDS-DC flow field with recirculation and Péclet-number diagnostics (original archived figure).

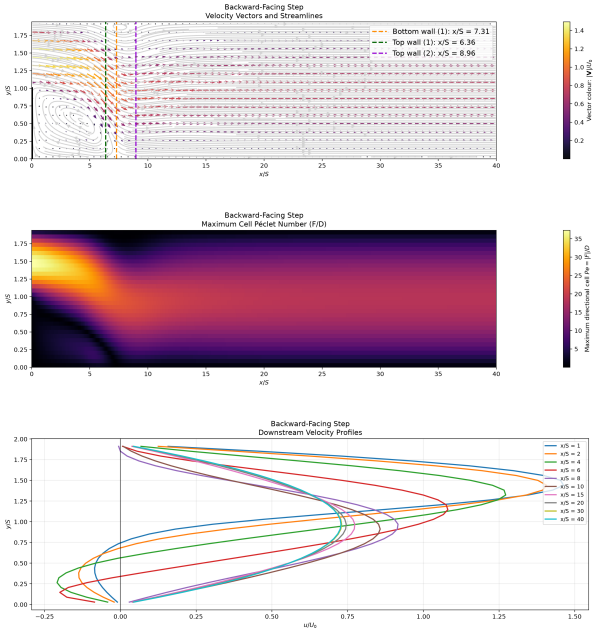
D2 :: Deferred correction

In deferred correction, the central convective flux is written as the upwind flux plus the difference between central and upwind fluxes. UDS coefficients are kept in the implicit matrix, while this flux difference is added as an explicit source evaluated using the latest iterate. At full convergence, the method recovers the CDS discrete convective operator, provided boundary fluxes are treated consistently. Deferred correction mainly changes the convergence path; changing from UDS to CDS changes the discretised equations.

Part D :: Additional convection-scheme plots

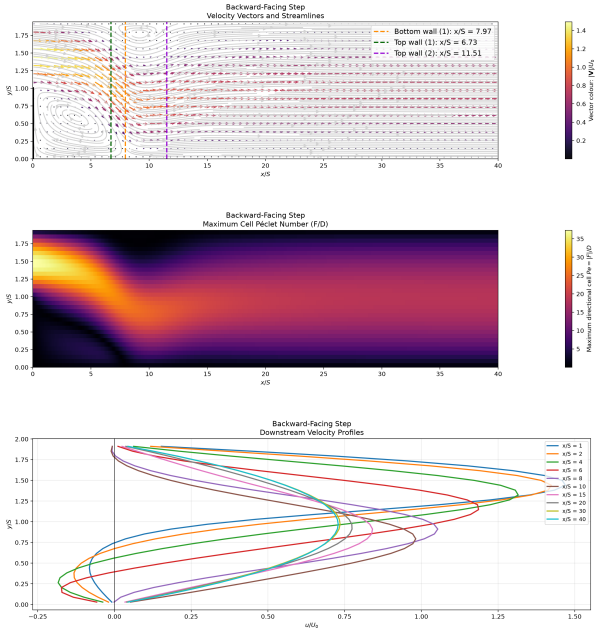
Grid dependence and Reynolds-number variation

D-V1 | Re = 400, UDS, 340×33



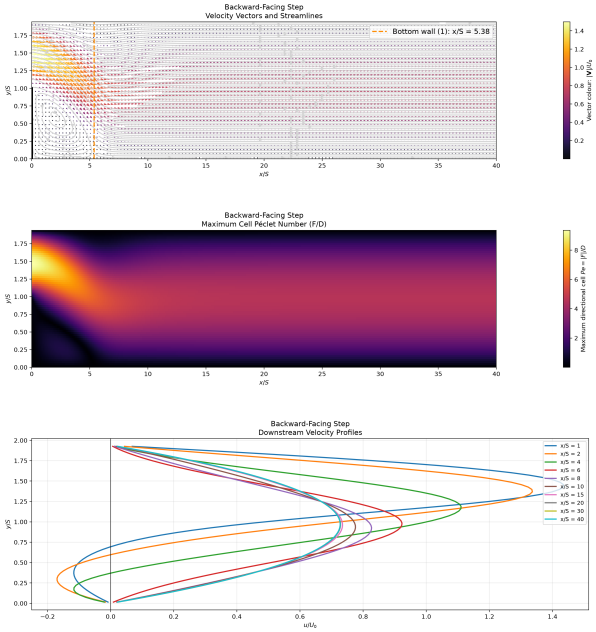
Coarse-grid UDS recirculation and velocity profiles.

D-V2 | Re = 400, CDS-DC, 340×33



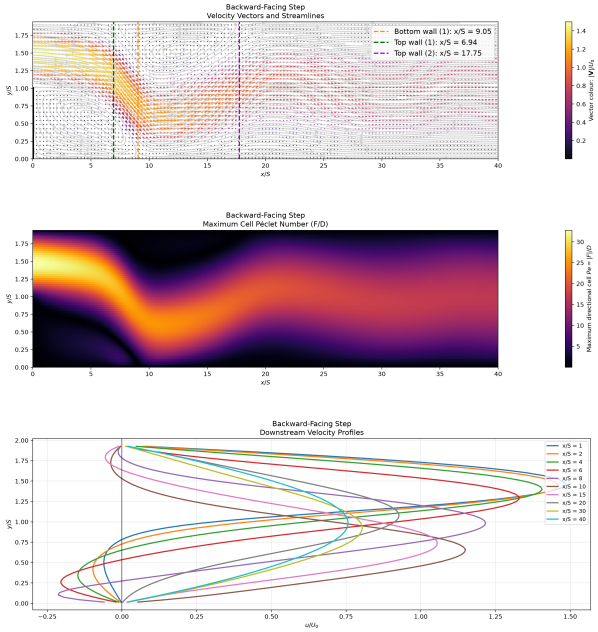
Same grid with deferred-correction convection.

D-V3 | Re = 200, CDS-DC, 680×66



Low-Re behaviour on the Reynolds-sweep grid.

D-V4 | Re = 700, CDS-DC, 680×66



Higher-Re upper-wall bubble on the same grid.

Part D3.1 :: Grid refinement and Richardson analysis

At Re = 400 and Δt = 0.05 s, the coarse grid has 340 × 33 cells and the fine grid has 640 × 66 cells. Wall-normal resolution is doubled, but streamwise resolution changes by a factor 640/340 ≈ 1.882. This does not form the exact 2:1 grid-refinement pair required in the question. The extrapolation is therefore not exact.

Scheme	340 × 33	640 × 66	Assumed p	Approx. $x_{1,\infty}/S$
UDS	7.30761	7.80034	1	8.32436
CDS-DC	7.96509	8.20925	2	8.29756

An effective planar spacing h_eff ∝ 1/sqrt(Nx Ny) gives r = √[(640×66)/(340×33)] = 1.9403. The illustrative Richardson estimate is x∞ ≈ xfine + (xfine − xcoarse)/(r^p − 1), using each scheme's formal order. Differences in cell aspect ratio, high cell Péclet numbers and finite-time error introduce uncertainty. The two extrapolated estimates differ by about 0.027S; this close agreement does not establish the observed convergence order.

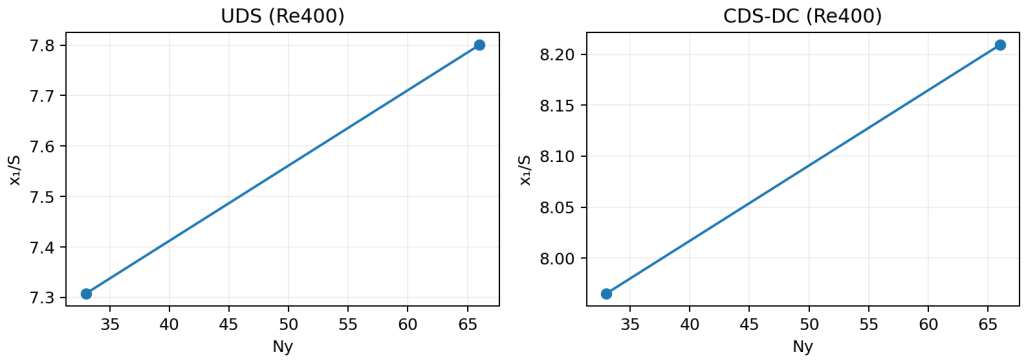


Fig. D3.1. Re = 400 coarse- and fine-grid reattachment lengths.

Numerical diffusion estimate

For the step geometry, h = 17/16, ν = 2h/Re = 2.125/Re. At Re = 400, ν = 0.0053125, so vnum/ν ≈ $U_0\Delta x/(2\nu)$ = (40/Nx)/(2ν). This is ≈11.07 for Nx = 340, ≈5.88 for Nx = 640 and ≈5.54 for Nx = 680. These large streamwise-Pe-based ratios explain the systematic shortening of the UDS recirculation zone relative to CDS-DC, although actual multidimensional false diffusion depends on flow orientation and gradients.

Courant number and discretisation error: Choosing Co = |U|Δt/Δx = 1 does not eliminate numerical diffusion in this implicit backward-Euler FVM solver. The cancellation at Co = 1 is a special result for explicit first-order upwind pure advection. For one-dimensional backward Euler with upwind convection, the leading numerical diffusion is approximately $|U|\Delta x/2 + U^2\Delta t/2$, which remains positive at Co = 1. Numerical dispersion is a separate error and does not automatically become large at Co = 1. Grid and time-step refinement, together with suitable bounded higher-order convection, are required to reduce these errors.

D3.3 :: Cost on the grid pair

Grid	Scheme	Mean iter./step	CPU (s)	x_1/S
340 × 33	UDS	12.09	1234.2	7.308
340 × 33	CDS-DC	12.91	1349.6	7.965
640 × 66	UDS	10.82	4896.4	7.800
640 × 66	CDS-DC	10.40	4756.6	8.209

For the available runs, CDS-DC takes about 9.4% more CPU time on the coarse grid and about 2.9% less on the fine grid. These are timings up to t = 100 s, not times to steady state, as neither fine-grid case reaches the prescribed stopping tolerance. A major reason for the high CPU cost is that the cell-Péclet chart was recalculated during every solver iteration. Repeated plotting adds considerable overhead. Without this repeated calculation and plotting, the runtime is expected to be much smaller, possibly by around two orders of magnitude. This reduction has not been independently measured, so the quoted CPU times should not be considered the cost of the numerical scheme alone. The cases with even finer grid took a maximum of 10 hrs to converge.

Part D3.2–D3.4 :: Reynolds-number dependence

The Reynolds-number sweep contains 680 × 66 runs at Re = 100, 200, 300, 500, 600 and 700 for CDS-DC. The Re = 700 UDS log is incomplete. Re = 400 results are reused from the 640 × 66 grid study, while Re = 800 UDS data are taken from the 330 × 33 Part C runs. A completed Re = 800 CDS-DC result is not available as the run failed ocntinuously (system limitations). Substitute points are marked separately because they do not belong to a uniform-grid sweep.

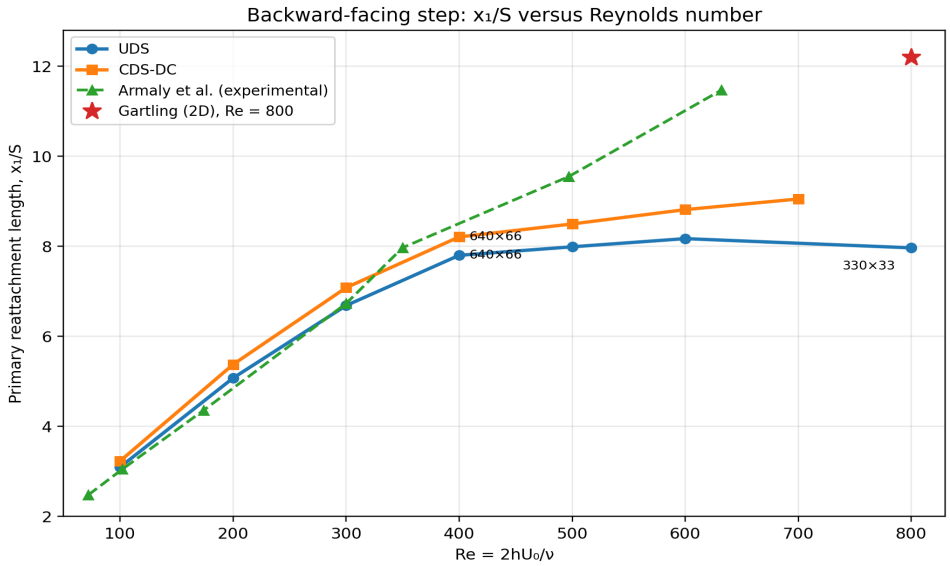


Fig. D3.2. Extracted x_1/S results; squares/triangle identify the nonmatching-grid substitute cases. Experimental comparison uses assignment-tabulated Armaly data and the assignment-cited Gartling benchmark.

Re	Grid	UDS x_1/S	CDS-DC x_1/S	CDS-DC x_4/S	CDS-DC x_5/S
100	680×66	3.0958	3.2312	—	—
200	680×66	5.0749	5.3762	—	—
300	680×66	6.6852	7.0821	—	—
400*	640×66	7.8003	8.2092	7.6835	10.2337
500	680×66	7.9893	8.4970	6.7777	14.3773
600	680×66	8.1708	8.8152	6.8543	16.5675
700	680×66	9.051	9.0515	6.9356	17.7475
800*	330×33 (680x66 failed)	7.9661	—	—	—

*Different-grid substituted results are reference points only. At Re = 800, the Part C UDS Δt = 0.05 case was used; x_4/S = 6.1489 and x_5/S = 13.5078 for that UDS run, but they are not CDS-DC sweep values.

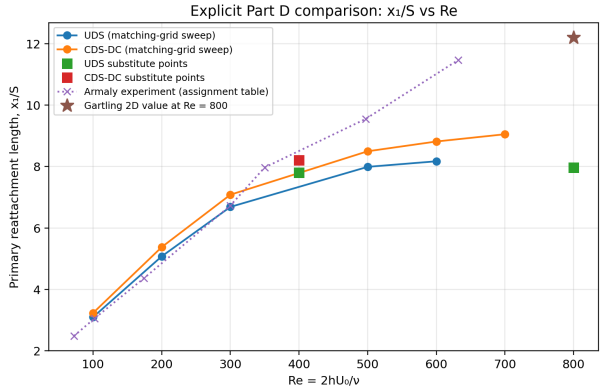
Comparison with Armaly and modelling limitations

Up to about Re = 300, the computed primary reattachment length follows the experimental trend, although the values are not identical. Beyond Re ≈ 400, the experimental channel develops three-dimensional flow, whereas these simulations are two-dimensional. Further differences arise because a parabolic inlet profile is imposed directly at the step, unlike the experimental upstream development. UDS false diffusion, grid resolution, incomplete steady convergence and the inconsistent grids also affect the numerical results. The high-Re discrepancy therefore cannot be attributed only to discretisation error or only to the 2D model.

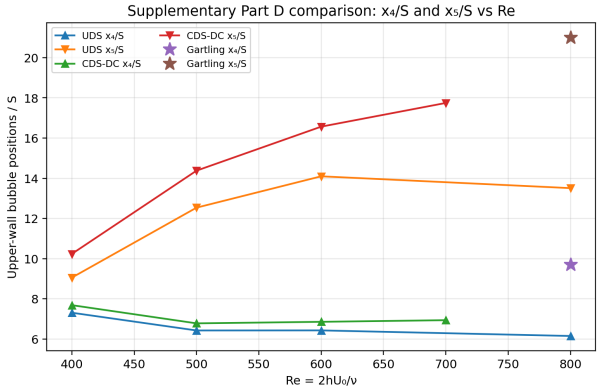
Armaly's data and computational solutions do no match beyond Re = 400 because : the numerical diffusion introduced by first-order upwinding (UDS). At higher Reynolds numbers, physical viscosity decreases, while the relative contribution of numerical viscosity becomes considerably larger. This causes excessive smoothing of velocity gradients, dissipation of local maxima and minima, and shortening of the recirculation region. Although CDS-DC reduces this numerical diffusion, discretisation errors still remain. Additionally, Armaly's experiments show three-dimensional flow beyond Re ≈ 400, whereas the present simulations assume two-dimensional flow, contributing further to the deviation.

Extra Part D plots

The x_1/S versus Re plot compares the numerical results with Armaly's values given in the assignment and Gartling's 2D reference at Re = 800. A companion plot shows x_4/S and x_5/S. Square markers identify reused results from different grid sizes and must not be interpreted as part of a consistent-grid study.



Supplementary Fig. D-S1. Explicit x_1/S comparison with Armaly experiment and Gartling 2D reference.



Supplementary Fig. D-S2. Upper-wall bubble limits x_4/S and x_5/S across Reynolds number.